Universidad del Desarrollo

Strategic Marketing

Certamen 2 · Nov 2026

InfoCar



B2B & B2C · For used-car dealers in Chile

Our plan to learn what certainty is worth.

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Dynamic: Information asymmetry

The seller knows the real car's condition.

SELLER

Knows the car

Past crashes, repaint, hidden filler, old faults.

?

Information gap

Condition is hard to check before paying

BUYER

Sees a price

An ad, a few photos, a short test drive.

RESULT

Every car is discounted

Doubt sets the price, so good cars leave the market.

What the gap costs

2.0–68.9M

CLP asking prices for the same model, the Ford F-150, in 2025. Price alone does not reveal condition.

Guzmán, 2026

52.3 days

average days from an insurance claim to the repaired car (2nd half 2025). Hidden damage is costly to fix.

Farías, 2026

Hidden condition = a cost on every deal

Concept: market for lemons (Akerlof, 1970). Sources: Guzmán (2026), Emol; Farías (2026), Autocosmos.

Big market, AI pricing arriving, but no Chilean dealer tool reads damage from photos

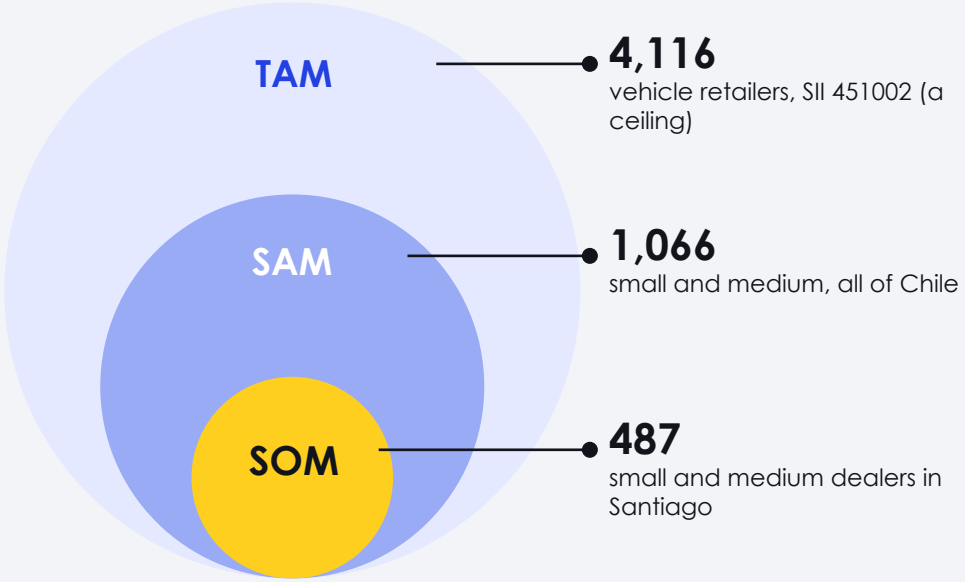
1,060,919
used-car transfers in 2025, up 5.8% · CAVEM (2026a)

15.3%
bought by companies, Sep 2026 · CAVEM (2026c)

58%
of transfers are in the Santiago Region · CAVEM (2026b)

3.4
used cars sold per new one · CAVEM (2026a)

Who we serve first SII, activity 451002



4,116
vehicle retailers, SII 451002 (a ceiling)

1,066
small and medium, all of Chile

487
small and medium dealers in Santiago

Competition

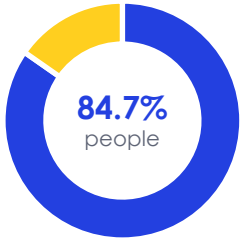
		Condition from photos
Rodar	Max purchase price from comparables and OBD2 readings. Closest to us.	✕
GoAuto, AutoGest	Market pricing for dealers. No photo-based condition, no fault database.	✕
Autofact, Carvuk, own eye	Legal history, in-person inspection or spreadsheets.	✕
InfoCar Where we stand out: damage read from photos, plus a fault database by model and salvage auctions. No hardware; works on any car, even one seen in a listing.		✓

Sources: CAVEM (2026a, 2026b, 2026c); SII (2026): activity 451002, year 2025, and list of firms in Santiago, 2024 (includes new-car sellers; large groups and micro-businesses out of focus); competitor websites (Rodar, n.d.; GoAuto, n.d.; AutoGest, n.d.; Carvuk, n.d.), checked 26 Sep 2026.

Who carries the gap of information

Who faces it

Share of used-car buyers, Sep 2026



People 84.7%

Companies 15.3%

Most buyers are people, but few buy often.

How often they buy

Private buyer one car at a time

Average dealer ≈3 cars a month (half buy ≈1)

Our scope: 487 dealers in Santiago

What widens it

Chinese brands, share of sales

New cars · Jan–Sep 2026



34.2%

Used cars · Jan–Sep 2026

12.95%

They reach the used market with little local record of typical faults or part prices.

Who protects the buyer?

Legal cover by type of seller

✓ **From a dealer**
No legal warranty if told the car is used (art. 14); consumer law still applies.

✕ **From a private seller**
No SERNAC protection: the buyer carries the risk.

41,700 **fraud cases in court, all types, Jan–Sep 2025, up 64%**
Melelli Viveros, 2026

Dealers face the gap on every purchase, new brands widen it, and the law barely covers it.

Sources: CAVEM (2026a, 2026c); SII (2026), purchases per dealer = sales ÷ average asking price; Ley N.° 19.496 (1997), art. 14; SERNAC (2019); Melelli Viveros (2026).

What a dealer pays to know

Every way to check a car today is slow, expensive, or blind to its physical condition.

Pre-purchase inspection

82–102k

CLP per car

1–3 hours. A mechanic checks the car in person.

Serviteca CyT, Ojo Mecánico

Chileautos home inspection

59,990

CLP per car + travel

130+ points, digital report in about an hour.

Chileautos

Autofact report

6–9k

CLP per car (5,990–8,990)

Price and legal history. Nothing on physical condition.

Autofact

Portals + spreadsheets

Hours

of staff time per car

Each salesperson prices the same car differently.

Our estimate, to test in SO4

Working price hypothesis: about CLP 59,900 a month, below one mechanic inspection

Sources: Serviteca CyT (n.d.), Ojo Mecánico (n.d.), Chileautos (n.d.), Autofact (n.d.-a, n.d.-b); prices checked 26 Oct and 7 Nov 2026.

Your own mechanic in your pocket



01 **Snap the car**
Guided photos, plus make, model, year, mileage and asking price.

02 **InfoCar does the maths**
Market comparables + the model's typical faults (validated by mechanics) + damage detected in the photos.

03 **Get a clear call**
Maximum purchase price, repair-cost range, and Buy / Negotiate / Pass.

Max price = market value − repairs − risk margin

A pre-screen, not a certificate: photos can't see inside the engine or gearbox. The price comes from a regression, not from the language model.

So… how to close the gap

Two revenue lines: the dealer pays a plan, and a private buyer pays per report.

INFOCAR

Condition report

Damage and repair cost, built from phone photos.

One report, two ways to pay.

- Visual appraisal with AI
- Parametric workshop budget
- Maximum purchase offer

B2B · THE DEALER PAYS

Prices with the repair cost known

Monthly plan · CLP 59,900–149,900

B2C · THE BUYER PAYS

Buys a car from a private seller

Where the law protects least.

One report · CLP 7,990–24,990

Today vs. InfoCar	Mechanic inspection	Rodar	InfoCar
Cost per car	**CLP 82,000–102,000**	**Not public**	**≈CLP 20,000 (average dealer)**
Time per car	1–3 hours	Needs a scanner fitted	Minutes (goal: under 5)
What it checks	Engine, body, chassis	Engine data (OBD-II)	Exterior body only

Sources: Serviteca CyT; Ojo Mecánico; Rodar (n.d.). InfoCar per car = CLP 59,900 ÷ ≈3 cars a month (SII, 2026). Prices: hypotheses tested in SO5.

Why not just ask ChatGPT?

A general chatbot has no current Chilean listings, and when it reads car damage from photos it often invents things.

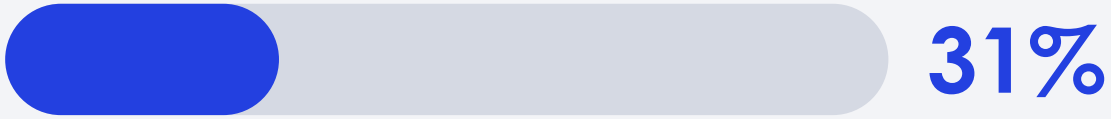
Vehicle damage reports with hallucinations

General vision model, image only



78%

Same model anchored to a dedicated damage-segmentation model



31%

100 human-checked reports (Hogale et al., 2026)

What InfoCar adds

- ✓ Up-to-date Chilean market data
- ✓ Vision models built for car damage
- ✓ Ranges with a confidence level
- ✓ Traceable estimates and human review

Accuracy is the top AI barrier for 74% of US franchise dealers (Cox Automotive, 2025): sell verifiable accuracy, not "AI"

Sources: Hogale et al. (2026), arXiv 2608.02470; Cox Automotive (2025).

What we need to understand to make it real

General objective: to understand how technical and economic certainty is valued when setting the transaction value of used cars.

Specific objective (SO) one dimension each	S Specific What exactly	M Measurable How we measure it	A Achievable Why it is doable	R Relevant Decision it informs	T Time-bound Weeks 1–4 (ideal)
1 **SO1 · Physical & aesthetic** **Identify** the physical and aesthetic attributes that determine how a car's condition is valued.	classification of body damages: dents, scratches, panel gaps, lights, repaint	Severity on a Likert 1–5 scale (1 = barely visible, 5 = structural)	Dealers judge these damages by eye every day	Sets the damage classes the app must detect	
2 **SO2 · Cost foresight** **Analyze** how anticipating reconditioning costs affects how the purchase offer is set.	CLP discount on the offer for each diagnosed damage	Choice experiment: blind case vs. case with repair cost	Pricing a trade-in is a daily task for appraisers	Calibrates the offer rule and the dealer's margin	
3 **SO3 · Adoption barriers** **Evaluate** the distrust barriers, resistance to automated diagnosis and technical uncertainty that shape transactions.	Fear of AI errors, of Chinese brands and of opaque appraisals	Coded objections + agreement on a Likert 1–5 scale	Objections come up on their own in interviews	Shapes how the report explains itself	
4 **SO4 · Workshop experience** **Describe** the operating experience and time bottlenecks of conventional exterior inspection and repair costing.	Days a car is stopped; hours to quote body and paint work	Time metrics (days, hours) + journey map	A known pain: 52.3 days from claim to repaired car	Sets the app flow and the time limit per report	
5 **SO5 · Appeal and adoption** **Evaluate** the perceived appeal, differentiation and adoption intent of a digital condition report vs. traditional appraisal.	B2B intent to subscribe; B2C intent to buy a report	Top-2 Box intent + Van Westendorp price meter	A clear stimulus: a mock-up of the digital report	Sets B2B and B2C prices before building the software	

T = theoretical plan, all five SOs within one month: W1 explore (stage 1) · W2 build and pilot the questionnaire · W3–W4 measure (stage 2).

Listen first, then measure

A mixed, sequential design: what we hear in stage 1 becomes the questions and scales of stage 2.

Stage 1

Qualitative, to explore

In-depth interviews (B2B) · Focus groups (B2C)

We learn the dealers' words, their rules of thumb for pricing, and what makes them distrust.

Explores SO1–SO5: the words and reasons

Bridge

Findings become variables, scales and closed answers.

Stage 2

Quantitative, to conclude

F2F + CATI (B2B) · CAWI + intercepts (B2C)

We measure the impact of each variable that we found, test our hypotheses, and estimate willingness to pay.

Measures SO1–SO5: how each variable impacts

Qualitative: one technique per customer

Interviews for what dealers keep private; groups for what buyers share.



In-depth interviews, one-on-one

B2B

Mid-size dealer owners | Appraisers

Why it fits best
Margins and workshop deals are trade secrets. Nobody shares them in front of competitors.

What it gives us
Hidden pricing rules, how far they trust AI, what fails in today's tools.

Technique · Semi-structured interview guide, 30–35 min, one-on-one

Moderated focus groups

B2C

Bought a used car in the last 12 months | Plan to buy one

Why it fits best
Fear of fraud is social: hearing others' stories makes people talk.

What it gives us
Shared risk perceptions, the buyer's own words, first reaction to the report.

Technique · Moderator's discussion guide, 75–90 min

Quantitative: meet each customer

Dealers in person at work, buyers online at scale. Phone calls and intercepts close the gaps.

B2B

F2F in situ + CATI

How we reach them
At the dealer's workplace, drawn from the SII list of firms: dealers rarely answer web surveys. CATI reaches firms we cannot visit.

Technique
Structured questionnaire on a tablet, 8–10 min, with the report prototype.

What it measures
Pricing rules at scale, reaction to the prototype, intent to subscribe.

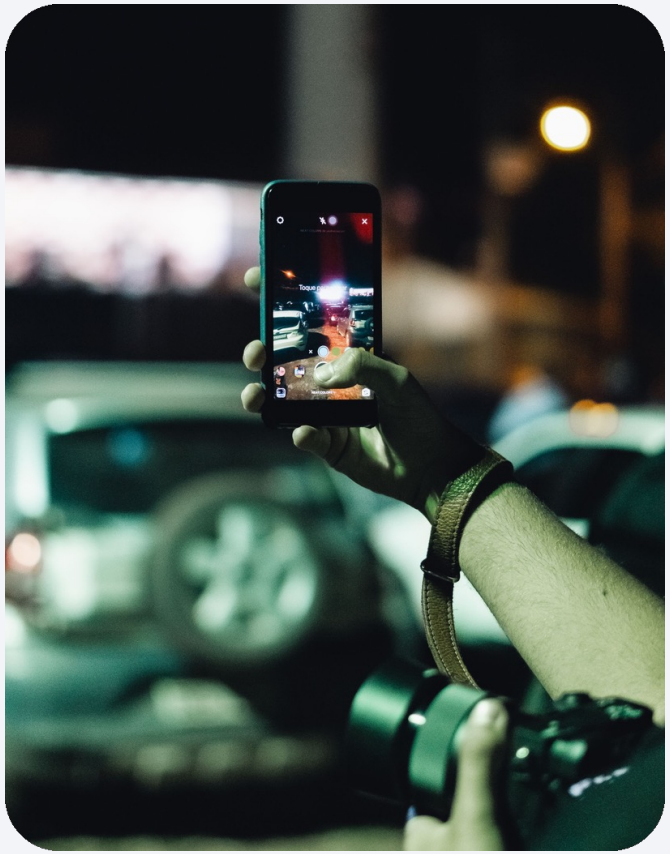
B2C

CAWI with quotas + intercepts

How we reach them
Online panel with quotas by age and purchase timing. Intercepts at notaries and PRT if a quota fills slowly.

Technique
Self-administered online questionnaire with the same report mock-up.

What it measures
How common each fear is, intent to buy a report, price range (PSM).



Each mode reaches one customer; together they cover both sides of the deal.

From method to objectives

Specific objective (SO)	Step 1 · Explore (qualitative)			Step 2 · Measure (quantitative)		Step 3 · Decide
	In-depth interviews B2B	Focus groups B2C	becomes questions	F2F in situ + CATI B2B	CAWI + intercepts B2C	Business decision
1 SO1 Physical & aesthetic	Words for each damage	Impact in the buying decision	→	Severity of kinds of damages, Likert 1–5	Most relevant aspects, Likert 1–5	Damage classes in the app
2 SO2 Cost foresight	Hidden costs	—		Choice experiment: blind vs. costed	—	Offer rule and margin
3 SO3 Adoption barriers	Objections to AI	Fear of fraud	→	Criticality, Likert 1–5	How common each fear is	How the report explains itself
4 SO4 Workshop experience	Where time is lost	—	→	Days stopped, hours to quote	—	App flow and timing
5 SO5 Appeal and adoption	First reaction to the prototype		→	Concept test, Top-2 Box	Intent + PSM	B2B and B2C prices

Who we will ask, and how many

Population and sample: the last link of the chain that starts with information asymmetry.

Diagnosis
Information asymmetry

→

Objective
What certainty is worth

→

Method
Listen first, then measure

→

After feedback

Population
Who belongs, and who does not

→

Sample
How many cases, and which

Five rules we follow: from class, checked against the literature

Five dimensions
Demographic, geographic, psychographic, consumption and need.

Filter, describe, segment
A filter decides who enters. A description says who they are. Only segments get cases.

30+ per cell
The floor to read a cell. Prices and totals are read on bigger bases.

Diversity first
Stage 1 looks for different profiles, not for large numbers.

Ideal, then realistic
Only a census covers everyone. We show the ideal and what fits in two weeks.

Filters: who gets in, and who does not

A filter is a yes-or-no rule. Everyone in the study passes all five of their market.

B2B Dealers

- ✓ Vehicle retailer, compraventas included (SII 451002)
- ✓ Small or medium: 2,400–100,000 UF a year
- ✓ Based in the Santiago Region
- ✓ Bought a used car to resell, last 3 months
- ✓ The person who answers sets the price

How many pass

4,116
retailers in Chile

1,066
small and medium

487
in Santiago, with name and tax ID

Out: large firms (89% of sales), micro firms, importers (451001), other regions (42% of transfers).

SII (2026), company statistics 2025 and list of legal persons 2024; CAVEM (2026b)

B2C Private buyers

- ✓ 18 or older
- ✓ Lives in the Santiago Region
- ✓ Bought a used car, SUV or pickup in the last 12 months, or will in the next 6
- ✓ Pays with own or household money
- ✓ Takes part in the decision

How many pass

1,060,919
transfers in Chile, 2025

615,311
in Santiago (58.0%)

≈521,000
bought by people (84.7%)

≈485,000
cars, SUVs, pickups (93%)

Out: companies, under 18, other regions, other vehicle types. Future buyers: no official count, the screener measures them.

Our estimate; CAVEM (2026b); CAVEM (2026c); SERNAC (2019)

Description: who they are, once inside

No yes-or-no here. We ask and analyse these; nobody is excluded or given a quota for them.

	B2B · dealers	B2C · private buyers
Demographic Always required	Informant's role, years in the trade, franchise or independent	Sex, NSE (AIM 2023, anonymous), occupation
Geographic Always required	Comuna of the business	Comuna of residence
Consumption SO2, SO5	Channels, tools paid for, share of used cars, mechanic on staff	Who sold (private or company), car value by permit, cash or credit
Need SO1, SO4	How often a car is priced without a full inspection	Checked the car before paying? With whom?
Psychographic SO3	Trust in AI, risk aversion, trust in own eye	Fear of fraud, trust in technology

Five dimensions: Scherman (2026a, 2026b). NSE: AIM (2024). Age is not here: it decides the cells, on the next slide.

Segments: only these get their own cases

Dealers split by the size of the firm; buyers by their age and the moment they buy.

B2B **Firm size, by SII sales bracket**

B1 · Pequeña 1–2 ≈1 car a month	**255** firms
B2 · Pequeña 3 ≈3 cars a month	**136** firms
B3 · Mediana ≈9 cars a month	**96** firms

Why size: the tax list gives it for every firm, so we can draw at random.

SII (2026); cars a month: sales ÷ asking price, Guzmán (2026)

B2C **Buyer's age × purchase timing**

	Bought, last 12 months	Will buy, next 6 months
18–29 first car	50 ideal · 30 real	50 ideal · 30 real
30–54 family, work	50 ideal · 30 real	50 ideal · 30 real
55+ renewal	50 ideal · 30 real	50 ideal · 30 real

Why age: bought on credit
Kavak sales, H1 2026



One platform, so a signal, not the market.

Seller type moves to description: we still record it, but age now decides the cells.

G5 Noticias (2026); Chócale (2026); age bands defined in the population, as seen in class

Car value cuts across both markets: next slide

We group cars by the value of their permit

Every car pays a circulation permit, set by law from the SII appraisal before any sale: objective and public.

How we set the cuts

- ✕ **Equal thirds**
  28% pay the legal minimum, so the first cut falls on it.
- ✕ **Legal tax brackets**
  Too unequal: 64% / 26% / 10%.
- ✓ **Dalenius–Hodges**
  The classic rule for strata of a skewed variable, on the log scale.
- ✓ **Fisher–Jenks check**
  A second method lands on nearly the same cuts.

Cuts at 100 and 200 UTM of appraisal so they hold every year. The three groups explain 86% of the variance in (log) permit value.

Three value segments

Share of used-car transfers, 2025

Low	**permit ≤ CLP 97,653** typical car: 2011	**55.7%**
Mid	**CLP 97,654–292,958** typical car: 2019	**28.0%**
High	**permit > CLP 292,958** typical car: 2023	**16.3%**

How we classify each car: brand, model and year → its permit in the SII table. Nobody has to remember what they paid.

Sources: SII (2026), appraisal and permit table 2026, 65,812 versions; weighted with CAVEM (2026b) transfers by brand and age; Decreto Ley 3.063, art. 12; Dalenius & Hodges (1959). UTM = CLP 69,752.

Stage 1: diversity, not numbers

Ideal design. Two interviews per dealer profile; one focus group per age band and purchase timing.

In-depth interviews · B2B 5 profiles × 2 = 10, plus 2 in reserve

1	**Small dealer** Pequeña · ≈1–3 cars a month	One with low-value stock, one with mid-value.	2 interviews
2	**Mid-size dealer** Mediana · ≈9 cars a month	One with an appraiser, one where the owner prices.	2 interviews
3	**Flipper** buys, fixes, resells	One buys crashed cars, one buys from people.	2 interviews
4	**High-value dealer** stock mostly in the high segment	One independent, one on consignment.	2 interviews
5	**Franchise appraiser** contrast, not target	One traditional brand, one Chinese brand.	2 interviews

Focus groups · B2C 3 × 2 = 6 groups

	Bought, last 12 months	Will buy, next 6 months
18–29	**FG1** 6–8 people	**FG2** 6–8 people
30–54	**FG3** 6–8 people	**FG4** 6–8 people
55+	**FG5** 6–8 people	**FG6** 6–8 people

Inside each group

8 recruited · low, mid and high car value · mixed sex, NSE and seller type · nobody who knows another.

Stop rule: 2 interviews in a row with no new theme. Saturation: 9–17 interviews, 4–8 groups (Hennink & Kaiser, 2022). Class: 2 interviews per profile; 2 groups per segment is rigorous (Scherman, 2026a, 2026b).

Ideal: 100 dealer surveys in three cells

A random draw from the SII list of Santiago, cell by cell, with replacements.

n = 100
of 487 dealers, one informant each

±9.1%
reference error for the total, 95%, after weighting and the finite-population correction

≈180–290 firms to contact
firm surveys get 34–55% answers (Holtom et al., 2022)

B1 Pequeña 1–2

40 of 255

16% of the cell ·±14.3%
weight 1.31 for totals

B2 Pequeña 3

30 of 136

22% of the cell ·±15.9%
weight 0.93 for totals

B3 Mediana

30 of 96

31% of the cell ·±14.9%
weight 0.66 for totals

Why 40 / 30 / 30: a proportional split gives 52 / 28 / 20, so B2 and B3 fall under the 30-case floor. They go up to 30; the other 40 go to B1. Weights bring totals back to the real mix.

Choice experiment · SO2
3 car cards per dealer, one per value segment, priced blind or with the repair cost, rotated.

300 appraisals

100 per segment

50 / 50 blind vs. with cost

Error = 1.96·√(0.25/n)·√((N−n)/(N−1)); design effect 1.07. Firm-level response: 34% (2015) and 55% (2020) in 1,014 published surveys (Holtom et al., 2022). B3 means contacting ≈90 of its 96 firms.

Ideal: 300 buyer surveys in six cells

Age × purchase timing, 50 per cell. Car value and seller type are recorded, not quota'd.

	Bought, last 12 months	Will buy, next 6 months
18–29 100 cases	50	50
30–54 100 cases	50	50
55+ 100 cases	50	50

Why 50: above the 30 floor in all six cells, and 300 is what the budget reaches.

How far each reading goes

Total	300	±5.7%
Per timing	150	±8.0% · prices read here
Per age band	100	±9.8% · prices read here
Per cell	50	±13.9%

≥220 online panel, quotas by age and timing

≤80 intercepts at notaries, a separate sub-sample

56/28/16 value weights

Reference error 1.96·√(0.25/n), 95%; quota samples have no true margin of error (AAPOR, Baker et al., 2013). No public source gives buyers' age mix, so the screener measures it and we do not weight by age. Value weights: permit segments.

Ideal vs. realistic: what fits in two weeks

Ideal = meets our precision targets with no time limit. Realistic = what seven of us can run in two weeks.

	Ideal (slides 19–21)	Realistic · two weeks	What we lose
Why	**30+ per cell and totals near ±10% or better**	**Seven people, two weeks, no paid panel**	Precision
Dealer interviews	10 + 2 reserve, 5 profiles	**8 + 2 reserve, 4 profiles, 30-min video calls**	The franchise contrast
Focus groups	6: age × timing	**3 online, 1 per age band, 5–6 people**	The timing split in groups
Dealer survey	100 · 3 SII cells · ±9.1%	**60 · 2 strata of 30 · ±14.1%**	Small vs. mid-size only
Buyer survey	300 · 6 cells of 50 · ±5.7%	**180 · 6 cells of 30 · ±7.3%**	Cells: ±14% → ±18%
Choice experiment	300 appraisals, 100 per segment	**180 appraisals, 60 per segment**	Small effects go unseen

Both designs in Santiago (487 dealers). Realistic: B2B weighted, design effect 1.37; B2C before weighting. Class: 3 groups per segment is ideal, 1 the realistic minimum (Scherman, 2026a).

Every objective has its cell

Who answers each specific objective, where we read it, and the trade-offs behind the numbers.

	Specific objective (SO)	Who answers	Where we read it	Decision
1	SO1 **Physical & aesthetic**	B2B B2C	Severity of each damage, by car value segment	Damage classes
2	SO2 **Cost foresight**	B2B	Appraisals blind vs. with cost, by value segment	Offer rule
3	SO3 **Adoption barriers**	B2B B2C	By age band; bought vs. about to buy	Report wording
4	SO4 **Workshop experience**	B2B	Days and hours lost, by dealer size (SII bracket)	App flow and timing
5	SO5 **Appeal and adoption**	B2B B2C	Prices: dealer total (100); each age band (100)	B2B and B2C prices

Budget · time · reach
what we compared

✓ Ideal B2B 100 | ✓ Real B2B 60 | ✕ B2B 150 | ✓ Ideal B2C 300 | ✓ Real B2C 180 | ✕ B2C 400

150 dealers would be 31% of the 487 in Santiago; 400 buyers needs a large paid panel. Ideal: 100 + 300. Realistic, two weeks: 60 + 180 (previous slide).

Two markets, one used car

Same car, same doubt. A dealer buys it to resell; a person buys it to live with.

B2B **Dealers who buy to resell**

487

small and medium dealers in Santiago

How often	≈3 cars a month on average; half buy ≈1
Why they buy	To resell at a margin
What they pay	Monthly plan, ≈CLP 59,900 (hypothesis)
Their cover	Sell under consumer law, art. 14

Guzmán (2026); SII (2026)

1,060,919

used cars changed hands in Chile in 2025



58%

of them in the Santiago Region

CAVEM (2026a); CAVEM (2026b)

B2C **People who buy for themselves**

≈485,000

purchases a year by people in Santiago

How often	One car at a time
Why they buy	To use it every day
What they pay	Per report, CLP 7,990–24,990
Their cover	From a private seller, none

CAVEM (2026b); CAVEM (2026c); SII (2026)

One information gap, lived from the two sides of the deal

What the data says, and what it cannot

Public statistics describe both markets. The gaps decide what we have to go and ask.

	B2B Dealers	B2C Private buyers
We know Public data	How many firms, and how big: 4,116 vehicle retailers, 1,066 small and medium, 487 in Santiago. SII (2026)	How many cars and who holds the title: 84.7% bought by people; 57% men and 30% women. CAVEM (2026c); 24 Horas (2024)
Nobody publishes The gap	How many used cars each dealer really buys (we can only estimate ≈3 a month), or how a price is set. Search: SII, CAVEM, CNC	How old used-car buyers are, or who sold to them. Only one platform reports age. G5 Noticias (2026); search: CAVEM, INE, SERNAC
So we ask Information need	Dealers, by firm size: how they price, what repairs cost them, what they would pay to know. SO2, SO4, SO5	Buyers, by age and purchase timing: what they fear, what they check, what a report is worth. SO1, SO3, SO5

The macro-environment: a PESTEL radiography

Everything around the used-car trade in Chile. Described, not judged.

PESTEL
The macro environment

→

Porter
The industry · next stage

→

SWOT
Comes after both



Political
Government, bills, trade

Economic
Growth, jobs, prices, credit

Social
Who buys, fraud, AI use

Technological
SMEs, data, damage AI

Environmental
Targets, fleet, batteries

Legal
Consumer, registry, data

The five rules of this radiography

Described, not judged
No finding is labelled opportunity or threat.

Wide and deep
Country, industry and global levels.

Source beside the idea
At least two per finding where they exist.

Both sides
Each finding keeps the data that cuts against it.

Space for doubt
Every letter closes with what is still open.

A finding without a source is dropped

How we built the radiography: every number was looked for in the document it cites.

160+
documents downloaded and read

88
findings kept in the research files

7
research runs: one per letter, one across

Not every source weighs the same

1. **Official data and law**
   Banco Central, INE, CMF, SII, LeyChile, Senado, MMA
2. **Global institutions**
   IEA, IMF, European Union, USTR
3. **Trade bodies**
   CAVEM, ANAC, CNC: data used; opinions marked as positions
4. **Press and platforms**
   A route to the data, always named as such

Corrected or dropped

"70% buy on credit"
A blog with no method. Dropped.

41.4% Chinese brands
Was country of manufacture. Now 34.2%.

41,700 "vehicle" frauds
Were all fraud types. Relabelled.

Full list of sources in the references. Not verified at source = not in this deck.

Political: the State reaches the car

A new government, a security agenda, a fuel decision, and rules still moving.

P E S T E L

2026–30

Country · Government

Kast government, in office since March 2026

The 2025–2031 security policy names vehicle theft and receiving stolen goods among its target crimes.

→ *A car's origin stays on the agenda across governments.*

Minsegpres (2026); Decreto N.° 96 (2026)

−22%

Industry · Car theft

violent car theft in 2025 (−2,293 cases)

−18.3% more in Q1 2026. All car thefts: 39,166 → 36,868. Prosecutors count 11,043 violent cases, the lowest since 2020.

→ *Theft is why buyers check plates, VIN and records.*

SPD (2026a); SPD (2026b); UNAB (2026); Fiscalía de Chile (2026)

−14 pts

Country · Fuel

presidential approval in three weeks (Cadem)

March 2026: the government loosened the MEPCO fuel stabiliser, at about USD 200M a week. Prices rose up to 54%; 60% called it avoidable.

→ *Fuel enters what a car costs to run.*

Emol (2026a); Emol (2026b); La República (2026)

New only

Industry · Bills

cars Chile may import today

Bill 18.607-15 (Aug 2026), from five senators, would allow some used EVs and hybrids. It is in the Senate Transport Committee.

→ *If passed, a new source of used cars.*

Ley N.° 18.483 (1985), art. 21; Senado (2026b)

12.5%

Global · Trade

US tariff on Chilean goods since 24 July 2026

It replaced a 10% temporary rate and reaches 60 economies. Passenger vehicles are among the general exclusions.

→ *Chile imports every new car: trade and the dollar set reference prices.*

Carey (2026); Emol (2026c)

Still open

Will Chile open to used imports?

Today: Law 18.483, art. 21: only unused vehicles may be imported.

The proposal: An exception for used EVs and hybrids. No vote has been held.

Ley N.° 18.483 (1985), art. 21; Senado (2026b)

Car crime falls, origin and fuel became political decisions, and none of the bills is law yet

Economic: used cars keep their own cycle

Growth under 1% and more people without a job, but more transfers and higher asking prices.

P E S T E L

0.25–0.75% — Country · Growth

GDP forecast for 2026, Central Bank, September

Cut from 1.0–1.75% in June. IMF, October: 0.6%. Both expect a recovery in 2027 (2–3%).

→ Frames household income and the cost of a car loan.

Banco Central de Chile (2026a); Vargas (2026); Pauta (2026)

+0.2% — Country · Activity

Imacec in September, after two months of decline

July and August fell (August −1.0%), pulled down by mining. Excluding mining, August grew 1.4%.

→ National figures hide very different sectors.

Banco Central de Chile (2026b); Meganoticias (2026)

9.6% — Country · Households

unemployment, June–August 2026 (+1.0 pt)

90-day bank arrears 2.36%; consumer loans fell 1.56% in September. Against a year ago, credit risk indicators are lower.

→ Jobs and credit decide who can finance a car.

INE (2026b); CMF (2026)

908,310 — Industry · Volume

used cars transferred, Jan–Oct 2026 (+3.0%)

3.4 used cars per new one. But October fell 7.0%.

→ The size of the market dealers work in.

CAVEM (2026a); The Clinic (2026b)

50% — Industry · Credit

of Kavak's sales financed, H1 2026 (34% in 2025)

One platform: the market-wide share is unknown.

→ Payment changes with the buyer's age.

G5 Noticias (2026); Chócale (2026)

Still open

What do used cars really sell for?

What is public: Asking prices: +5% in 2025, CLP 18.4M on average.

What is not: An index of real sale prices. In the US, wholesale fell 0.6% while used EVs rose 4.3%.

Guzmán (2026); Cox Automotive (2026b)

The economy slows in 2026 while transfers and asking prices rise; real sale prices remain unobserved

Social: who buys, and how they decide

Mostly people, an ageing country, fraud at a 12-year high; AI to compare, eyes to decide.

P E **S** T E L

85.7% — Industry · Who buys

of used cars bought by people, October 2026

84.7% in September. In the last split published by sex: 57% men, 30% women, 14% firms (Oct 2024).

→ *The figure says who buys, not who sold.*

The Clinic (2026b); CAVEM (2026c); 24 Horas (2024)

19.8% — Country · Age

of residents are 60 or older, Census 2024

Santiago Region: 1,392,757 people. Class B licence procedures, 2023: 30–39 lead with 24.7%, then 40–49 (21.5%) and 50–59 (17.5%).

→ *Licences are the closest public proxy for drivers' age.*

INE (2026a); INE (2025a)

41,700 — Country · Fraud

fraud cases in court, Jan–Sep 2025 (+64%)

All fraud types, a 12-year high; 69% are scams. No official figure isolates car fraud. Prosecutors: 68% of all cases have no known suspect.

→ *Why buyers pay for ownership and lien checks.*

Melelli Viveros (2026); Ministerio Público (2026)

78% — Country · AI use

of Chileans have used generative AI

43% to compare prices, but 26% before buying (56% read reviews). 59% say they cannot spot AI errors.

→ *High use, low confidence in judging it.*

The Clinic (2026a); Diario USACH (2026)

86% — Global · In person

of US consumers need to see the car

53% did every step at the dealer; 7% bought fully online.

→ *Physical inspection stays in the purchase.*

Deloitte (2025); Cox Automotive (2026a)

Still open

Does age change how people buy?

A signal: On Kavak, 58% of buyers aged 18–23 use credit; 56% of those 55+ pay cash.

The gap: No official data on the age of used-car buyers, or on who sold.

G5 Noticias (2026); Chócale (2026); CAVEM (2026c)

People are most of the market; AI enters the comparison, not the decision; the buyer's age is unmeasured

Technological: connected, but early

Phones and data are everywhere; daily use is slower, and photo AI has reported limits.

P E S **T** E L

70%

Country · SMEs

of SMEs say they use AI (large firms: over 80%)

But 70% of 40,000+ self-assessed small firms sit at an initial digital level. Lack of training is a main barrier (40%).

→ Using AI is not building it into daily work.

Entel Digital (2025); Min. de Economía (2026)

10.8M

Country · Connectivity

5G connections, June 2026 (+31.8%)

96.5% of households have internet. 5G reaches 51.9% of households, 37.9% in rural areas.

→ Photos and records can be checked on the lot.

Subtel (2026a); Subtel (2026b)

CLP 8,990

Industry · Car data

a full vehicle history report

Owners, mileage, liens and fines in one report. A State portal checks theft records. The official transfer is still in person.

→ Data on a car comes from several sources.

Autofact (n.d.-a); SPD (2026c); ChileAtiende (2026)

55

Global · Damage AI

studies on AI damage detection reviewed

Results vary by task and data. Minor damage and uncontrolled photos stay hard.

→ Photo AI reads visible damage, within limits.

Hasan et al. (2025); Lin et al. (2024)

14%

Global · Dealers

of US dealers have AI built into their workflow

74% cite accuracy as a barrier. 82% say they use AI; 22% saw the sales lift expected.

→ Reported use grows faster than integration.

Cox Automotive (2025); Cox Automotive (2026c)

Still open

What can a photo really prove?

What works: Models classify and locate visible damage.

What does not: A photo cannot certify the engine, and language models tend to guess when unsure.

Hasan et al. (2025); Kalai et al. (2025)

Connectivity and AI use are broad; integration is early, and how far a photo can go is still open

Environmental: new cars electrify

Targets apply to new cars; the used fleet lags by years; abroad, batteries enter the price.

P E S T E L

2035 — Country · Targets

100% zero-emission new light vehicles (20% by 2030)

Plug-ins were 6.3% of new sales in Jan–Sep 2026; EVs and hybrids are 0.5% of the fleet (2024).

→ *Flow changes fast, stock changes slowly.*

Min. de Energía (2026); ANAC (2026a); INE (2025b)

23.8% — Industry · Fleet age

of 2025 transfers were cars 16 or more years old

A fleet of 6.74M vehicles that renews 4.9% a year.

→ *Cars of very different standards share the market.*

CAVEM (2026b); INE (2025b)

≤ 2011 — Country · Santiago

green-sticker cars restricted in winter 2026

Cars with no green sticker cannot enter the Américo Vespucio ring. In 2018 such cars lost 10% of value in five months.

→ *Model year decides where a car may drive.*

MMA (2026); CAVEM (2018)

35% — Global · Used EVs

value kept by used EVs in Europe, 2025 (market: ≈50%)

In the US, used EV retail prices were 8% above a year earlier in Q3 2026.

→ *Regions move in different directions.*

IEA (2026); Recurrent (2026)

2.3% — Global · Batteries

average battery capacity lost per year (22,700 EVs)

The EU requires a battery passport from February 2027.

→ *Battery health becomes part of the price.*

Geotab (2026); European Parliament & Council (2023)

Still open

Will 800 models leave in 2027?

CAVEM's position: The 2027 efficiency rule (22.8 km/lge) would push more than 800 models out.

The rule itself: The standard is an average per importer, not a limit per model.

CAVEM (2026e); Min. de Energía (2022)

New sales electrify while the used market stays old; abroad, battery data starts to shape value

Legal: what is set, and what is moving

The rules in force for a used-car sale, and a data law whose date is being voted on.

P E S T E L

Art. 14

Country · Consumer law

Law 19.496 on consumer protection

If a dealer says a car is used, the legal warranty does not apply; any warranty it offers does. A private seller is not a 'provider'.

→ *Who sells decides what cover exists.*

Ley N.° 19.496 (1997); SERNAC (2019)

2023

Country · Registry

Law 21.601 against the sale of stolen cars

Plates etched on windows and mirrors. First-sale invoices only from issuers on a Civil Registry list.

→ *More checks before a car is registered.*

Ley N.° 21.601 (2023); SPD (2026c)

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Industry · Warranties

JAC X200 under a SERNAC safety alert, Feb 2026

Warranties differ on resale: MG's battery cover passes to new owners; GWM's 10-year engine cover is for the first owner only.

→ *Remaining cover depends on brand and owner.*

SERNAC (2026); MG Motor Chile (n.d.); GWM Chile (n.d.)

1 Dec 2026

Country · Data law

Law 21.719 takes effect, by law today

Fines up to 5,000, 10,000 or 20,000 UTM by severity. 72% of 127 organisations consulted said they were not ready.

→ *Apps that process photos and buyer data fall under it.*

Ley N.° 21.719 (2024); Cámara de Diputadas y Diputados (2026)

Air Canada

Global · AI liability

a firm answers for what its chatbot says (2024)

In Chile, an AI framework bill is in the Senate; the Executive proposed a redesign in May 2026.

→ *Who answers for an AI's output is being defined.*

American Bar Association (2024); Senado (2026a)

Still open

2026 or 2027?

Moving it: Senate 33–6 on 28 Oct; a Chamber committee agreed unanimously on 5 Nov: 15 Oct 2027.

Its critics: Some deputies fear repeated delays. Until published, 1 Dec 2026 stands.

Contreras Espinoza (2026); Cámara de Diputadas y Diputados (2026); Portal Innova (2026)

Disclosure, registry checks and warranties are set; the date of the data law is not settled

Across PESTEL: Chinese brands (1/2)

One factor, seen from several letters: economic, political, social.

P E S T E L

Economic · Share

34.2%

of new cars, Jan–Sep 2026, were Chinese brands

In used cars: 12.95% (95,353). By country of manufacture, China reaches 47.2% of new cars (Jan–Aug).

→ Their used stock grows as recent cohorts reach resale.

CAVEM (2026c); ANAC (2026b); The Clinic (2026b)

Political · Trade

0.49%

average effective tariff under the Chile–China FTA

Elsewhere: US 100% on Chinese EVs (2024), EU duties 17.0–35.3%. Chile stays open while large markets close.

→ Where Chinese cars go is decided abroad.

SUBREI (2025); USTR (2024); European Commission (2024)

Social · Perception

48%

would never buy a car made in China

Yet 49% link Chinese brands with value for money. In China, reliability: 201 problems per 100 cars vs 191 for international brands.

→ Origin and attributes pull in opposite directions.

Ortega (2025); J.D. Power (2026)

Chinese brands are a third of new sales and 13% of used ones, with mixed perceptions

Across PESTEL: Chinese brands (2/2)

China's support is shifting. What does that do to a used car's value in Chile?

P E S T E L

Global · State support

Half

of the purchase-tax exemption since January 2026

The national EV subsidy ended in December 2022. The tax break is halved for 2026–27, capped at CNY 15,000; the IEA expects more price pressure.

→ *New-car prices are the reference for used ones.*

IEA (2025); IEA (2026); Benchmark (2025); CN Biz Insight (2026)

Economic · Resale

35–45%

value lost in 3 years, Chinese brands (others: 25–35%)

An expert estimate, in the press. Chileautos puts the gap at 5–10 points.

→ *Depreciation varies by brand, not just origin.*

Gutiérrez (2025); CAVEM (2026d)

Still open

What happens to used values?

One view
Chinese brands depreciate 5–10 points more.

Another
CAVEM: brands with 10+ years in Chile keep competitive values. No public resale curve exists.

Gutiérrez (2025); CAVEM (2026d)

Support in China is changing and resale values differ by brand; the effect on Chile is unmeasured

The PESTEL canvas

Every letter on one page: what we see, key figures, what is still open. Selection comes next.

P Political

Car crime falls; fuel price and traceability became political decisions

−22%
violent car theft in 2025 (−2,293 cases)
SPD (2026a); SPD (2026b); UNAB (2026); Fiscalía de Chile (2026)

12.5%
US tariff on Chilean goods since 24 July 2026
Carey (2026); Emol (2026c)

Still open
A bill could open Chile to used EV imports

E Economic

Growth under 1% and more unemployment, while used-car transfers rise

0.25–0.75%
GDP forecast for 2026, Central Bank, September
Banco Central de Chile (2026a); Vargas (2026); Pauta (2026)

908,310
used cars transferred, Jan–Oct 2026 (+3.0%)
CAVEM (2026a); The Clinic (2026b)

Still open
No public index of real sale prices

S Social

People buy most used cars, compare with AI and still want to see them

85.7%
of used cars bought by people, October 2026
The Clinic (2026b); CAVEM (2026c); 24 Horas (2024)

78%
of Chileans have used generative AI
The Clinic (2026a); Diario USACH (2026)

Still open
No official data on buyers' age or seller

T Technological

Wide connectivity, shallow AI use; photo AI works within known limits

10.8M
5G connections, June 2026 (+31.8%)
Subtel (2026a); Subtel (2026b)

70%
of SMEs say they use AI (large firms: over 80%)
Entel Digital (2025); Min. de Economía (2026)

Still open
What a photo can really prove

E Environmental

New sales electrify; the fleet stays old; batteries enter the price abroad

2035
100% zero-emission new light vehicles (20% by 2030)
Min. de Energía (2026); ANAC (2026a); INE (2025b)

23.8%
of 2025 transfers were cars 16 or more years old
CAVEM (2026b); INE (2025b)

Still open
Whether 800 models leave in 2027

L Legal

Disclosure and registry rules are set; data and AI rules are moving

Art. 14
Law 19.496 on consumer protection
Ley N.° 19.496 (1997); SERNAC (2019)

1 Dec 2026
Law 21.719 takes effect, by law today
Ley N.° 21.719 (2024); Cámara de Diputadas y Diputados (2026)

Still open
Data law: Dec 2026 or Oct 2027

Across letters Chinese brands: a third of new sales, 13% of used, valued brand by brand

Still open: what lower support in China does to resale

A full radiography, on purpose: selection starts with Porter and SWOT

Universidad del Desarrollo

Thank you

1 Diagnosis
Information asymmetry between buyers and sellers.

2 Objective
Understand what certainty is worth.

3 Method
Listen first, then measure: 100 + 300 surveys, 60 + 180 in two weeks.

4 Environment
A PESTEL radiography: described, not judged.

Questions?



What changed after your feedback

Appendix. What we heard, what we did, and where to find it.

What we heard	What we changed	Slides
Segment by age more than by seller type	B2C cells are now age × purchase timing; seller type is recorded	17, 19, 21–23
Give B2C the same weight as B2B	Both markets side by side, in every step of the story	14–17, 24–25
Separate filters from descriptions	One slide for filters, one for descriptions, one for segments	15–17
Start with a general view of both markets	One slide with B2B and B2C, before the macro-environment	24
Hard data, checked at the source	Fraud, Chinese share, repair days and plan price corrected	2–5, 7–9
Look at the Chinese wave and the outside world	Two cross-cutting slides and a global level in each letter	28–35

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